DBMS ASSIGNMENT

Names: Fabrice Mukunzi

Student ID: 29620

1. Concatenate first and last name as full_name.

```
MariaDB [employee_management]> SELECT CONCAT(first_name, ' ', last_name) AS full_name FROM employees;
+---------------+
| full_name     |
+---------------+
| Alice Johnson |
| Bob Smith     |
| Carol Adams   |
| David Lee     |
| Eve Martins   |
| Frank Green   |
| Grace Brown   |
| Hank Wilson   |
| Ivy Clark     |
| Jake White    |
+---------------+
10 rows in set (0.017 sec)

MariaDB [employee_management]>
```

2. Convert all employee names to lowercase.

```
MariaDB [employee_management]> SELECT LOWER(first_name) AS lower_first, LOWER(last_name) AS lower_last FROM employees;
+-------------+------------+
| lower_first | lower_last |
+-------------+------------+
| alice       | johnson    |
| bob         | smith      |
| carol       | adams      |
| david       | lee        |
| eve         | martins    |
| frank       | green      |
| grace       | brown      |
| hank        | wilson     |
| ivy         | clark      |
| jake        | white      |
+-------------+------------+
10 rows in set (0.015 sec)

MariaDB [employee_management]>
```

3. Extract first 3 letters of the employee's first name.

```
MariaDB [employee_management]> SELECT SUBSTRING(first_name, 1, 3) AS first3 FROM employees;
+--------+
| first3 |
+--------+
| Ali    |
| Bob    |
| Car    |
| Dav    |
| Eve    |
| Fra    |
| Gra    |
| Han    |
| Ivy    |
| Jak    |
+--------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

4. Replace '@company.com' in email with '@org.com'

```
MariaDB [employee_management]> SELECT REPLACE(email, '@company.com', '@org.com') AS updated_email FROM employees;
+-----------------------+
| updated_email         |
+-----------------------+
| alice.johnson@org.com |
| bob.smith@org.com     |
| carol.adams@org.com   |
| david.lee@org.com     |
| eve.martins@org.com   |
| frank.green@org.com   |
| grace.brown@org.com   |
| hank.wilson@org.com   |
| ivy.clark@org.com     |
| jake.white@org.com    |
+-----------------------+
10 rows in set (0.015 sec)

MariaDB [employee_management]>
```

5. Trim spaces from a padded string.

```
MariaDB [employee_management]> SELECT TRIM('   Hello World   ') AS trimmed_string;
+----------------+
| trimmed_string |
+----------------+
| Hello World    |
+----------------+
1 row in set (0.002 sec)

MariaDB [employee_management]>
```

6. Count characters in an employee’s full name.

```
MariaDB [employee_management]> SELECT LENGTH(CONCAT(first_name, ' ', last_name)) AS name_length FROM employees;
+-------------+
| name_length |
+-------------+
|          13 |
|           9 |
|          11 |
|           9 |
|          11 |
|          11 |
|          11 |
|          11 |
|           9 |
|          10 |
+-------------+
10 rows in set (0.002 sec)

MariaDB [employee_management]>
```

7. Find position of '@' in email using INSTR()/CHARINDEX().

```
MariaDB [employee_management]> SELECT INSTR(email, '@') AS at_position FROM employees;
+-------------+
| at_position |
+-------------+
|          14 |
|          10 |
|          12 |
|          10 |
|          12 |
|          12 |
|          12 |
|          12 |
|          10 |
|          11 |
+-------------+
10 rows in set (0.014 sec)

MariaDB [employee_management]>
```

8. Add ‘Mr.’ or ‘Ms.’ before names based on gender (assume gender exists).

```
MariaDB [employee_management]> SELECT
    ->   CASE
    ->     WHEN gender = 'M' THEN CONCAT('Mr. ', first_name, ' ', last_name)
    ->     WHEN gender = 'F' THEN CONCAT('Ms. ', first_name, ' ', last_name)
    ->     ELSE CONCAT(first_name, ' ', last_name)
    ->   END AS titled_name
    -> FROM employees;
ERROR 1054 (42S22): Unknown column 'gender' in 'SELECT'
MariaDB [employee_management]>
```

9. Format project names to uppercase.

```
MariaDB [employee_management]> SELECT UPPER(project_name) AS upper_name FROM projects;
+---------------------------+
| upper_name                |
+---------------------------+
| HR REVAMP                 |
| FINANCE AUTOMATION        |
| IT INFRASTRUCTURE UPGRADE |
| MARKETING BLITZ 2025      |
| LEGAL COMPLIANCE          |
| CUSTOMER PORTAL           |
| SALES BOOSTER             |
| R&D PILOT                 |
| PROCUREMENT TRACKER       |
| OPERATIONS STREAMLINE     |
+---------------------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

10. Remove any dashes from project names.

```
MariaDB [employee_management]> SELECT REPLACE(project_name, '-', '') AS cleaned_name FROM projects;
+---------------------------+
| cleaned_name              |
+---------------------------+
| HR Revamp                 |
| Finance Automation        |
| IT Infrastructure Upgrade |
| Marketing Blitz 2025      |
| Legal Compliance          |
| Customer Portal           |
| Sales Booster             |
| R&D Pilot                 |
| Procurement Tracker       |
| Operations Streamline     |
+---------------------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

11. Create a label like “Emp: John Doe (HR)”.

```
MariaDB [employee_management]> SELECT CONCAT('Emp: ', first_name, ' ', last_name, ' (', department_name, ')') AS label
    -> FROM employees
    -> JOIN departments USING (department_id);
+-------------------------------------------+
| label                                     |
+-------------------------------------------+
| Emp: Alice Johnson (Human Resources)      |
| Emp: Bob Smith (Information Technology)   |
| Emp: Carol Adams (Finance)                |
| Emp: David Lee (Marketing)                |
| Emp: Eve Martins (Information Technology) |
| Emp: Frank Green (Sales)                  |
| Emp: Grace Brown (Legal)                  |
| Emp: Hank Wilson (Operations)             |
| Emp: Ivy Clark (Research and Development) |
| Emp: Jake White (Customer Service)        |
+-------------------------------------------+
10 rows in set (0.020 sec)

MariaDB [employee_management]>
```

12. Check email length for each employee.

```
MariaDB [employee_management]> SELECT email, LENGTH(email) AS email_length FROM employees;
+---------------------------+--------------+
| email                     | email_length |
+---------------------------+--------------+
| alice.johnson@company.com |           25 |
| bob.smith@company.com     |           21 |
| carol.adams@company.com   |           23 |
| david.lee@company.com     |           21 |
| eve.martins@company.com   |           23 |
| frank.green@company.com   |           23 |
| grace.brown@company.com   |           23 |
| hank.wilson@company.com   |           23 |
| ivy.clark@company.com     |           21 |
| jake.white@company.com    |           22 |
+---------------------------+--------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

13. Extract last name only from email (before @).

```
MariaDB [employee_management]> SELECT SUBSTRING_INDEX(SUBSTRING_INDEX(email, '@', 1), '.', -1) AS last_name_from_email F
ROM employees;
+----------------------+
| last_name_from_email |
+----------------------+
| johnson              |
| smith                |
| adams                |
| lee                  |
| martins              |
| green                |
| brown                |
| wilson               |
| clark                |
| white                |
+----------------------+
10 rows in set (0.015 sec)

MariaDB [employee_management]>
```

14. Format: “LASTNAME, Firstname” using UPPER and CONCAT

```
MariaDB [employee_management]> SELECT CONCAT(UPPER(last_name), ', ', first_name) AS formatted_name FROM employees;
+----------------+
| formatted_name |
+----------------+
| JOHNSON, Alice |
| SMITH, Bob     |
| ADAMS, Carol   |
| LEE, David     |
| MARTINS, Eve   |
| GREEN, Frank   |
| BROWN, Grace   |
| WILSON, Hank   |
| CLARK, Ivy     |
| WHITE, Jake    |
+----------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

15. Add “(Active)” next to employee names who have current projects.

```
MariaDB [employee_management]> SELECT
    ->   CONCAT(first_name, ' ', last_name,
    ->     CASE
    ->       WHEN p.end_date IS NULL THEN ' (Active)'
    ->       ELSE ''
    ->     END
    ->   ) AS employee_status
    -> FROM employees e
    -> LEFT JOIN employee_projects ep ON e.employee_id = ep.employee_id
    -> LEFT JOIN projects p ON ep.project_id = p.project_id;
+----------------------+
| employee_status      |
+----------------------+
| Alice Johnson        |
| Bob Smith (Active)   |
| Carol Adams          |
| David Lee            |
| Eve Martins (Active) |
| Frank Green          |
| Grace Brown          |
| Hank Wilson          |
| Ivy Clark (Active)   |
| Jake White           |
+----------------------+
10 rows in set (0.018 sec)

MariaDB [employee_management]>
```

16. Round salary to the nearest whole number.

```
MariaDB [employee_management]> SELECT employee_id, ROUND(salary) AS rounded_salary FROM employees;
+-------------+----------------+
| employee_id | rounded_salary |
+-------------+----------------+
|         101 |           4500 |
|         102 |           5200 |
|         103 |           6700 |
|         104 |           3800 |
|         105 |           4000 |
|         106 |           6000 |
|         107 |           4900 |
|         108 |           3100 |
|         109 |           2700 |
|         110 |           3600 |
+-------------+----------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

17. Show only even salaries using MOD.

```
MariaDB [employee_management]> SELECT * FROM employees WHERE MOD(salary, 2) = 0;
+-------------+------------+-----------+---------------------------+------------+---------+---------------+
| employee_id | first_name | last_name | email                     | hire_date  | salary  | department_id |
+-------------+------------+-----------+---------------------------+------------+---------+---------------+
|         101 | Alice      | Johnson   | alice.johnson@company.com | 2015-03-15 | 4500.00 |             1 |
|         102 | Bob        | Smith     | bob.smith@company.com     | 2018-06-23 | 5200.00 |             3 |
|         103 | Carol      | Adams     | carol.adams@company.com   | 2012-09-10 | 6700.00 |             2 |
|         104 | David      | Lee       | david.lee@company.com     | 2020-01-05 | 3800.00 |             4 |
|         105 | Eve        | Martins   | eve.martins@company.com   | 2019-12-11 | 4000.00 |             3 |
|         106 | Frank      | Green     | frank.green@company.com   | 2017-07-08 | 6000.00 |             8 |
|         107 | Grace      | Brown     | grace.brown@company.com   | 2014-11-02 | 4900.00 |             5 |
|         108 | Hank       | Wilson    | hank.wilson@company.com   | 2013-02-17 | 3100.00 |             6 |
|         109 | Ivy        | Clark     | ivy.clark@company.com     | 2021-08-30 | 2700.00 |             9 |
|         110 | Jake       | White     | jake.white@company.com    | 2022-05-19 | 3600.00 |             7 |
+-------------+------------+-----------+---------------------------+------------+---------+---------------+
10 rows in set (0.002 sec)

MariaDB [employee_management]>
```

18. Show difference between two project end/start dates using DATEDIFF.

```
MariaDB [employee_management]> SELECT project_name, DATEDIFF(end_date, start_date) AS project_duration FROM projects WHE
RE end_date IS NOT NULL;
+----------------------+------------------+
| project_name         | project_duration |
+----------------------+------------------+
| HR Revamp            |              364 |
| Finance Automation   |              350 |
| Marketing Blitz 2025 |              149 |
| Legal Compliance     |              184 |
| Customer Portal      |              364 |
| Sales Booster        |              364 |
| Procurement Tracker  |              245 |
| Operations Streamline |             365 |
+----------------------+------------------+
8 rows in set (0.015 sec)

MariaDB [employee_management]>
```

19. Show absolute difference in salaries between two employees.

```
MariaDB [employee_management]> SELECT ABS(e1.salary - e2.salary) AS diff
    -> FROM employees e1, employees e2
    -> WHERE e1.employee_id = 101 AND e2.employee_id = 102;
+--------+
| diff   |
+--------+
| 700.00 |
+--------+
1 row in set (0.015 sec)

MariaDB [employee_management]>
```

20. Raise salary by 10% using POWER.

```
MariaDB [employee_management]> SELECT employee_id, salary, salary * POWER(1.10, 1) AS new_salary FROM employees;
+-------------+---------+--------------------+
| employee_id | salary  | new_salary         |
+-------------+---------+--------------------+
|         101 | 4500.00 |               4950 |
|         102 | 5200.00 |  5720.000000000001 |
|         103 | 6700.00 |  7370.000000000001 |
|         104 | 3800.00 |               4180 |
|         105 | 4000.00 |               4400 |
|         106 | 6000.00 |  6600.000000000001 |
|         107 | 4900.00 |               5390 |
|         108 | 3100.00 | 3410.0000000000005 |
|         109 | 2700.00 | 2970.0000000000005 |
|         110 | 3600.00 | 3960.0000000000005 |
+-------------+---------+--------------------+
10 rows in set (0.014 sec)

MariaDB [employee_management]>
```

21. Generate a random number for testing IDs.

```
MariaDB [employee_management]> SELECT RAND() AS random_num;
+--------------------+
| random_num         |
+--------------------+
| 0.6510300903125015 |
+--------------------+
1 row in set (0.014 sec)

MariaDB [employee_management]>
```

22. Use CEIL and FLOOR on a floating salary.

```
MariaDB [employee_management]> SELECT salary, CEIL(salary) AS ceil_val, FLOOR(salary) AS floor_val FROM employees;
+---------+----------+-----------+
| salary  | ceil_val | floor_val |
+---------+----------+-----------+
| 4500.00 |     4500 |      4500 |
| 5200.00 |     5200 |      5200 |
| 6700.00 |     6700 |      6700 |
| 3800.00 |     3800 |      3800 |
| 4000.00 |     4000 |      4000 |
| 6000.00 |     6000 |      6000 |
| 4900.00 |     4900 |      4900 |
| 3100.00 |     3100 |      3100 |
| 2700.00 |     2700 |      2700 |
| 3600.00 |     3600 |      3600 |
+---------+----------+-----------+
10 rows in set (0.002 sec)

MariaDB [employee_management]>
```

23. Use LENGTH() on phone numbers (assume column exists).

```
MariaDB [employee_management]> SELECT employee_id, phone_number, LENGTH(phone_number) AS phone_length FROM emplo
yees;
ERROR 1054 (42S22): Unknown column 'phone_number' in 'SELECT'
MariaDB [employee_management]>
```

24. Categorize salary: High/Medium/Low using CASE.

```
MariaDB [employee_management]> SELECT employee_id,
    ->   CASE
    ->     WHEN salary >= 5000 THEN 'High'
    ->     WHEN salary >= 3000 THEN 'Medium'
    ->     ELSE 'Low'
    ->   END AS salary_category
    -> FROM employees;
+-------------+-----------------+
| employee_id | salary_category |
+-------------+-----------------+
|         101 | Medium          |
|         102 | High            |
|         103 | High            |
|         104 | Medium          |
|         105 | Medium          |
|         106 | High            |
|         107 | Medium          |
|         108 | Medium          |
|         109 | Low             |
|         110 | Medium          |
+-------------+-----------------+
10 rows in set (0.008 sec)

MariaDB [employee_management]>
```

25. Count digits in salary amount.

```
MariaDB [employee_management]> SELECT salary, LENGTH(FLOOR(salary)) AS digit_count FROM employees;
+---------+-------------+
| salary  | digit_count |
+---------+-------------+
| 4500.00 |           4 |
| 5200.00 |           4 |
| 6700.00 |           4 |
| 3800.00 |           4 |
| 4000.00 |           4 |
| 6000.00 |           4 |
| 4900.00 |           4 |
| 3100.00 |           4 |
| 2700.00 |           4 |
| 3600.00 |           4 |
+---------+-------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

26. Show today's date using CURRENT_DATE.

```
MariaDB [employee_management]> SELECT CURRENT_DATE;
+--------------+
| CURRENT_DATE |
+--------------+
| 2025-07-31   |
+--------------+
1 row in set (0.029 sec)

MariaDB [employee_management]>
```

27. Calculate how many days an employee has worked.

```
MariaDB [employee_management]> SELECT employee_id, DATEDIFF(CURDATE(), hire_date) AS days_worked FROM employees;
+-------------+-------------+
| employee_id | days_worked |
+-------------+-------------+
|         101 |        3791 |
|         102 |        2595 |
|         103 |        4707 |
|         104 |        2034 |
|         105 |        2059 |
|         106 |        2945 |
|         107 |        3924 |
|         108 |        4547 |
|         109 |        1431 |
|         110 |        1169 |
+-------------+-------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

28. Show employees hired in the current year.

```
MariaDB [employee_management]> SELECT * FROM employees WHERE YEAR(hire_date) = YEAR(CURDATE());
Empty set (0.001 sec)

MariaDB [employee_management]>
```

29. Display current date and time using NOW().

```
MariaDB [employee_management]> SELECT NOW();
+---------------------+
| NOW()               |
+---------------------+
| 2025-07-31 07:55:05 |
+---------------------+
1 row in set (0.001 sec)

MariaDB [employee_management]>
```

30. Extract the year, month, and day from hire_date.

```
MariaDB [employee_management]> SELECT hire_date, YEAR(hire_date) AS hire_year, MONTH(hire_date) AS hire_month, DAY(hire_
date) AS hire_day FROM employees;
+------------+-----------+------------+----------+
| hire_date  | hire_year | hire_month | hire_day |
+------------+-----------+------------+----------+
| 2015-03-15 |      2015 |          3 |       15 |
| 2018-06-23 |      2018 |          6 |       23 |
| 2012-09-10 |      2012 |          9 |       10 |
| 2020-01-05 |      2020 |          1 |        5 |
| 2019-12-11 |      2019 |         12 |       11 |
| 2017-07-08 |      2017 |          7 |        8 |
| 2014-11-02 |      2014 |         11 |        2 |
| 2013-02-17 |      2013 |          2 |       17 |
| 2021-08-30 |      2021 |          8 |       30 |
| 2022-05-19 |      2022 |          5 |       19 |
+------------+-----------+------------+----------+
10 rows in set (0.008 sec)

MariaDB [employee_management]>
```

31. Show employees hired before 2020.

```
MariaDB [employee_management]> SELECT * FROM employees WHERE hire_date < '2020-01-01';
+-------------+------------+-----------+---------------------------+------------+---------+---------------+
| employee_id | first_name | last_name | email                     | hire_date  | salary  | department_id |
+-------------+------------+-----------+---------------------------+------------+---------+---------------+
|         101 | Alice      | Johnson   | alice.johnson@company.com | 2015-03-15 | 4500.00 |             1 |
|         102 | Bob        | Smith     | bob.smith@company.com     | 2018-06-23 | 5200.00 |             3 |
|         103 | Carol      | Adams     | carol.adams@company.com   | 2012-09-10 | 6700.00 |             2 |
|         105 | Eve        | Martins   | eve.martins@company.com   | 2019-12-11 | 4000.00 |             3 |
|         106 | Frank      | Green     | frank.green@company.com   | 2017-07-08 | 6000.00 |             8 |
|         107 | Grace      | Brown     | grace.brown@company.com   | 2014-11-02 | 4900.00 |             5 |
|         108 | Hank       | Wilson    | hank.wilson@company.com   | 2013-02-17 | 3100.00 |             6 |
+-------------+------------+-----------+---------------------------+------------+---------+---------------+
7 rows in set (0.008 sec)

MariaDB [employee_management]>
```

32. List projects that ended in the last 30 days.

```
MariaDB [employee_management]> SELECT * FROM projects
    -> WHERE end_date IS NOT NULL
    -> AND end_date BETWEEN DATE_SUB(CURDATE(), INTERVAL 30 DAY) AND CURDATE();
Empty set (0.002 sec)

MariaDB [employee_management]>
```

33. Calculate total days between project start and end dates.

```
MariaDB [employee_management]> SELECT project_name, DATEDIFF(end_date, start_date) AS total_days
    -> FROM projects
    -> WHERE end_date IS NOT NULL;
+----------------------+------------+
| project_name         | total_days |
+----------------------+------------+
| HR Revamp            |        364 |
| Finance Automation   |        350 |
| Marketing Blitz 2025 |        149 |
| Legal Compliance     |        184 |
| Customer Portal      |        364 |
| Sales Booster        |        364 |
| Procurement Tracker  |        245 |
| Operations Streamline |       365 |
+----------------------+------------+
8 rows in set (0.001 sec)

MariaDB [employee_management]>
```

34. Format date: ‘2025-07-23’ to ‘July 23, 2025’ (use CONCAT).

```
MariaDB [employee_management]> SELECT CONCAT(MONTHNAME('2025-07-23'), ' ', DAY('2025-07-23'), ', ', YEAR('2025-07-23'))
AS formatted_date;
+----------------+
| formatted_date |
+----------------+
| July 23, 2025  |
+----------------+
1 row in set (0.007 sec)

MariaDB [employee_management]>
```

35. Add a CASE: if project still active (end_date IS NULL), show 'Ongoing'.

```
MariaDB [employee_management]> SELECT project_name,
    ->   CASE
    ->     WHEN end_date IS NULL THEN 'Ongoing'
    ->     ELSE 'Completed'
    ->   END AS status
    -> FROM projects;
+---------------------------+-----------+
| project_name              | status    |
+---------------------------+-----------+
| HR Revamp                 | Completed |
| Finance Automation        | Completed |
| IT Infrastructure Upgrade | Ongoing   |
| Marketing Blitz 2025      | Completed |
| Legal Compliance          | Completed |
| Customer Portal           | Completed |
| Sales Booster             | Completed |
| R&D Pilot                 | Ongoing   |
| Procurement Tracker       | Completed |
| Operations Streamline     | Completed |
+---------------------------+-----------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

36. Use CASE to label salaries.

```
MariaDB [employee_management]> SELECT employee_id,
    ->   CASE
    ->     WHEN salary >= 5000 THEN 'High'
    ->     WHEN salary >= 3000 THEN 'Medium'
    ->     ELSE 'Low'
    ->   END AS salary_category
    -> FROM employees;
+-------------+-----------------+
| employee_id | salary_category |
+-------------+-----------------+
|         101 | Medium          |
|         102 | High            |
|         103 | High            |
|         104 | Medium          |
|         105 | Medium          |
|         106 | High            |
|         107 | Medium          |
|         108 | Medium          |
|         109 | Low             |
|         110 | Medium          |
+-------------+-----------------+
10 rows in set (0.008 sec)

MariaDB [employee_management]>
```

37. Use COALESCE to show 'No Email' if email is NULL.

```
MariaDB [employee_management]> SELECT COALESCE(email, 'No Email') AS email_display FROM employees;
+---------------------------+
| email_display             |
+---------------------------+
| alice.johnson@company.com |
| bob.smith@company.com     |
| carol.adams@company.com   |
| david.lee@company.com     |
| eve.martins@company.com   |
| frank.green@company.com   |
| grace.brown@company.com   |
| hank.wilson@company.com   |
| ivy.clark@company.com     |
| jake.white@company.com    |
+---------------------------+
10 rows in set (0.008 sec)

MariaDB [employee_management]>
```

38. CASE: If hire_date < 2015, mark as ‘Veteran’.

```
MariaDB [employee_management]> SELECT employee_id,
    ->   CASE WHEN hire_date < '2015-01-01' THEN 'Veteran'
    ->        ELSE 'New'
    ->   END AS service_status
    -> FROM employees;
+-------------+----------------+
| employee_id | service_status |
+-------------+----------------+
|         101 | New            |
|         102 | New            |
|         103 | Veteran        |
|         104 | New            |
|         105 | New            |
|         106 | New            |
|         107 | Veteran        |
|         108 | Veteran        |
|         109 | New            |
|         110 | New            |
+-------------+----------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

39. If salary is NULL, default it to 3000 using COALESCE.

```
MariaDB [employee_management]> SELECT COALESCE(salary, 3000) AS final_salary FROM employees;
+--------------+
| final_salary |
+--------------+
|      4500.00 |
|      5200.00 |
|      6700.00 |
|      3800.00 |
|      4000.00 |
|      6000.00 |
|      4900.00 |
|      3100.00 |
|      2700.00 |
|      3600.00 |
+--------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

40. CASE: Categorize departments (IT, HR, Other)

```
MariaDB [employee_management]> SELECT department_name,
    ->   CASE
    ->     WHEN department_name = 'Information Technology' THEN 'IT'
    ->     WHEN department_name = 'Human Resources' THEN 'HR'
    ->     ELSE 'Other'
    ->   END AS category
    -> FROM departments;
+--------------------------+----------+
| department_name          | category |
+--------------------------+----------+
| Human Resources          | HR       |
| Finance                  | Other    |
| Information Technology   | IT       |
| Marketing                | Other    |
| Legal                    | Other    |
| Operations               | Other    |
| Customer Service         | Other    |
| Sales                    | Other    |
| Research and Development | Other    |
| Procurement              | Other    |
+--------------------------+----------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

41. CASE: If employee has no project, mark as 'Unassigned'.

```
MariaDB [employee_management]> SELECT e.employee_id,
    ->   CASE
    ->     WHEN ep.project_id IS NULL THEN 'Unassigned'
    ->     ELSE 'Assigned'
    ->   END AS project_status
    -> FROM employees e
    -> LEFT JOIN employee_projects ep ON e.employee_id = ep.employee_id;
+-------------+----------------+
| employee_id | project_status |
+-------------+----------------+
|         101 | Assigned       |
|         103 | Assigned       |
|         102 | Assigned       |
|         105 | Assigned       |
|         104 | Assigned       |
|         107 | Assigned       |
|         108 | Assigned       |
|         110 | Assigned       |
|         106 | Assigned       |
|         109 | Assigned       |
+-------------+----------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

42. CASE: Show tax band based on salary.

```
MariaDB [employee_management]> SELECT employee_id, salary,
    ->   CASE
    ->     WHEN salary >= 6000 THEN 'High Tax'
    ->     WHEN salary >= 4000 THEN 'Medium Tax'
    ->     ELSE 'Low Tax'
    ->   END AS tax_band
    -> FROM employees;
+-------------+---------+------------+
| employee_id | salary  | tax_band   |
+-------------+---------+------------+
|         101 | 4500.00 | Medium Tax |
|         102 | 5200.00 | Medium Tax |
|         103 | 6700.00 | High Tax   |
|         104 | 3800.00 | Low Tax    |
|         105 | 4000.00 | Medium Tax |
|         106 | 6000.00 | High Tax   |
|         107 | 4900.00 | Medium Tax |
|         108 | 3100.00 | Low Tax    |
|         109 | 2700.00 | Low Tax    |
|         110 | 3600.00 | Low Tax    |
+-------------+---------+------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

43. Use nested CASE to label project duration.

```
MariaDB [employee_management]> SELECT project_name,
    ->   CASE
    ->     WHEN end_date IS NULL THEN 'Ongoing'
    ->     WHEN DATEDIFF(end_date, start_date) > 365 THEN 'Long-term'
    ->     ELSE 'Short-term'
    ->   END AS duration_label
    -> FROM projects;
+----------------------------+----------------+
| project_name               | duration_label |
+----------------------------+----------------+
| HR Revamp                  | Short-term     |
| Finance Automation         | Short-term     |
| IT Infrastructure Upgrade  | Ongoing        |
| Marketing Blitz 2025       | Short-term     |
| Legal Compliance           | Short-term     |
| Customer Portal            | Short-term     |
| Sales Booster              | Short-term     |
| R&D Pilot                  | Ongoing        |
| Procurement Tracker        | Short-term     |
| Operations Streamline      | Short-term     |
+----------------------------+----------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

44. Use CASE with MOD to show even/odd salary IDs.

```
MariaDB [employee_management]> SELECT employee_id,
    ->   CASE
    ->     WHEN MOD(employee_id, 2) = 0 THEN 'Even'
    ->     ELSE 'Odd'
    ->   END AS id_type
    -> FROM employees;
+-------------+---------+
| employee_id | id_type |
+-------------+---------+
|         101 | Odd     |
|         103 | Odd     |
|         102 | Even    |
|         105 | Odd     |
|         104 | Even    |
|         107 | Odd     |
|         108 | Even    |
|         110 | Even    |
|         106 | Even    |
|         109 | Odd     |
+-------------+---------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

45. Combine COALESCE + CONCAT for fallback names.

```
MariaDB [employee_management]> SELECT COALESCE(CONCAT(first_name, ' ', last_name), 'No Name') AS display_name FROM emplo
yees;
+---------------+
| display_name  |
+---------------+
| Alice Johnson |
| Bob Smith     |
| Carol Adams   |
| David Lee     |
| Eve Martins   |
| Frank Green   |
| Grace Brown   |
| Hank Wilson   |
| Ivy Clark     |
| Jake White    |
+---------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

46. CASE with LENGTH(): if name length > 10, label “Long Name”.

```
MariaDB [employee_management]> SELECT CONCAT(first_name, ' ', last_name) AS full_name,
    ->   CASE
    ->     WHEN LENGTH(CONCAT(first_name, last_name)) > 10 THEN 'Long Name'
    ->     ELSE 'Short Name'
    ->   END AS name_type
    -> FROM employees;
+---------------+------------+
| full_name     | name_type  |
+---------------+------------+
| Alice Johnson | Long Name  |
| Bob Smith     | Short Name |
| Carol Adams   | Short Name |
| David Lee     | Short Name |
| Eve Martins   | Short Name |
| Frank Green   | Short Name |
| Grace Brown   | Short Name |
| Hank Wilson   | Short Name |
| Ivy Clark     | Short Name |
| Jake White    | Short Name |
+---------------+------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

47. CASE + UPPER(): if email has 'TEST', mark as dummy account.

```
MariaDB [employee_management]> SELECT email,
    ->   CASE
    ->     WHEN UPPER(email) LIKE '%TEST%' THEN 'Dummy Account'
    ->     ELSE 'Valid'
    ->   END AS email_status
    -> FROM employees;
+---------------------------+--------------+
| email                     | email_status |
+---------------------------+--------------+
| alice.johnson@company.com | Valid        |
| bob.smith@company.com     | Valid        |
| carol.adams@company.com   | Valid        |
| david.lee@company.com     | Valid        |
| eve.martins@company.com   | Valid        |
| frank.green@company.com   | Valid        |
| grace.brown@company.com   | Valid        |
| hank.wilson@company.com   | Valid        |
| ivy.clark@company.com     | Valid        |
| jake.white@company.com    | Valid        |
+---------------------------+--------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

48. CASE: Show seniority based on hire year (e.g., Junior/Senior).

```
MariaDB [employee_management]> SELECT employee_id, hire_date,
    ->   CASE
    ->     WHEN YEAR(hire_date) <= 2015 THEN 'Senior'
    ->     WHEN YEAR(hire_date) > 2015 THEN 'Junior'
    ->     ELSE 'Unknown'
    ->   END AS seniority
    -> FROM employees;
+-------------+------------+-----------+
| employee_id | hire_date  | seniority |
+-------------+------------+-----------+
|         101 | 2015-03-15 | Senior    |
|         102 | 2018-06-23 | Junior    |
|         103 | 2012-09-10 | Senior    |
|         104 | 2020-01-05 | Junior    |
|         105 | 2019-12-11 | Junior    |
|         106 | 2017-07-08 | Junior    |
|         107 | 2014-11-02 | Senior    |
|         108 | 2013-02-17 | Senior    |
|         109 | 2021-08-30 | Junior    |
|         110 | 2022-05-19 | Junior    |
+-------------+------------+-----------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

49. Use CASE to determine salary increment range.

```
MariaDB [employee_management]> SELECT salary,
    ->   CASE
    ->     WHEN salary < 3000 THEN 'Raise by 20%'
    ->     WHEN salary < 5000 THEN 'Raise by 10%'
    ->     ELSE 'Raise by 5%'
    ->   END AS increment_range
    -> FROM employees;
+---------+-----------------+
| salary  | increment_range |
+---------+-----------------+
| 4500.00 | Raise by 10%    |
| 5200.00 | Raise by 5%     |
| 6700.00 | Raise by 5%     |
| 3800.00 | Raise by 10%    |
| 4000.00 | Raise by 10%    |
| 6000.00 | Raise by 5%     |
| 4900.00 | Raise by 10%    |
| 3100.00 | Raise by 10%    |
| 2700.00 | Raise by 20%    |
| 3600.00 | Raise by 10%    |
+---------+-----------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```

50. Use CASE with CURDATE() to determine anniversary month.

```
MariaDB [employee_management]> SELECT employee_id, hire_date,
    ->   CASE
    ->     WHEN MONTH(hire_date) = MONTH(CURDATE()) THEN 'Anniversary Month'
    ->     ELSE 'Not Anniversary'
    ->   END AS anniversary_status
    -> FROM employees;
+-------------+------------+--------------------+
| employee_id | hire_date  | anniversary_status |
+-------------+------------+--------------------+
|         101 | 2015-03-15 | Not Anniversary    |
|         102 | 2018-06-23 | Not Anniversary    |
|         103 | 2012-09-10 | Not Anniversary    |
|         104 | 2020-01-05 | Not Anniversary    |
|         105 | 2019-12-11 | Not Anniversary    |
|         106 | 2017-07-08 | Anniversary Month  |
|         107 | 2014-11-02 | Not Anniversary    |
|         108 | 2013-02-17 | Not Anniversary    |
|         109 | 2021-08-30 | Not Anniversary    |
|         110 | 2022-05-19 | Not Anniversary    |
+-------------+------------+--------------------+
10 rows in set (0.001 sec)

MariaDB [employee_management]>
```